Affine Dual Jacobi–Trudi Identities for Classical Hall–Littlewood Polynomials via Complementary Minors

Abstract

We prove the six affine dual Jacobi–Trudi identities conjectured by Warnaar for rectangular Hall–Littlewood polynomials of types B_n, C_n, and BC_n, together with his additional type-B_n identity at $s = t$. We give a uniform proof by adapting the complementary-minor method used by Aitken for symmetric-function duality.

Keywords. Hall–Littlewood polynomials; Jacobi–Trudi identities; Weyl characters; complementary minors; affine root systems.

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1 Introduction and main identities

Classical dual Jacobi–Trudi identities express rectangular Weyl characters of rank n as determinants of order k, where k is the width of the rectangle. Warnaar proposed affine analogues for classical Hall–Littlewood polynomials in [7, Conjectures 1.2–1.4], and a further type-B_n identity in [7, Section 6]. We prove these identities, with one necessary correction at width zero. Warnaar established the $k = 1$ cases of the six numbered conjectures and the $t = 1$ specializations (1.6a) and (1.8) [7, Lemmas 4.1–4.4]; his type-A analogue is [7, Theorem 1.1]. The applications to affine and q, t-Rogers–Ramanujan identities developed there use the classical-type formulas proved here.

For $x = (x_1, \dots, x_n)$, define

$$\begin{aligned} \dot{E}(u) &= \prod_{a=1}^{n} (1 + ux_a)(1 + ux_a^{-1}) = \sum_{r=0}^{2n} \dot{e}_r u^r, \\ \ddot{E}(u) &= (1 + u)\dot{E}(u) = \sum_{r=0}^{2n+1} \ddot{e}_r u^r. \end{aligned} \tag{1.1}$$

Both coefficient families are extended by zero to all integer indices. Thus

$$\dot{e}_{2n-r} = \dot{e}_r, \qquad \ddot{e}_{2n+1-r} = \ddot{e}_r, \qquad \ddot{e}_r = \dot{e}_r + \dot{e}_{r-1}.$$

For $1 \le i \le k$ and $m \in \mathbb{Z}$, use the row kernels:

$$\begin{aligned} \Lambda_i(m) &= \dot{e}_{n-i+m} - \dot{e}_{n+i+m}, & h_i(m) &= \dot{e}_{n-i+m} + \dot{e}_{n+i+m-1}, \\ h_i^+(m) &= \dot{e}_{n-i+m} + \dot{e}_{n+i+m-2}, & \widehat{h}_i(m) &= \ddot{e}_{n-i+m} + \ddot{e}_{n+i+m-1}. \end{aligned}$$

For weights $w_j(y)$, kernels $F_i(m)$, and $\mathcal{L} \subseteq \mathbb{Z}^k$, put

$$\mathfrak{J}_K[w, F; \mathcal{L}] := \sum_{(y_1,\dots,y_k)\in\mathcal{L}} \det_{1\le i,j\le k}[w_j(y_i)F_i(j - Ky_i)]. \tag{1.2}$$

In this notation an expression in j, y in the weight slot specifies $w_j(y)$. All sums below are finite by zero extension. Write $\mathbb{Z}^k_{\mathrm{even}} = \{y \in \mathbb{Z}^k : y_1 + \cdots + y_k \in 2\mathbb{Z}\}$. The empty determinant and the sum indexed by the empty vector both have value 1. When $k = 0$, no entry or weight in (1.2) is evaluated. The coefficient field is $\mathbb{Q}(t, x_1, \dots, x_n)$, except in the two square-root cases, where it is $\mathbb{Q}(\tau, x_1, \dots, x_n)$ and $t^{r/2} := \tau^r$ for $r \in \mathbb{Z}$.

Theorem A. *For $n \ge 1$, the following identities hold for $k \ge 0$, except that (1.5) requires $k \ge 1$:*

$$P^{B_n}_{(k^n)}(x; t, 0) = \mathfrak{J}_{2k+1}[(-1)^y t^{(k+\frac12)y^2-(j-\frac12)y}, h; \mathbb{Z}^k], \tag{1.3}$$

$$P^{B_n}_{(k^n)}(x; \tau^2, -\tau) = \mathfrak{J}_{2k}[\tau^{2ky^2-(2j-1)y}, h; \mathbb{Z}^k], \tag{1.4}$$

$$P^{B_n}_{(k^n)}(x; t, 1) = \tfrac12\mathfrak{J}_{2k}[(-1)^y t^{ky^2-(j-1)y}, h^+; \mathbb{Z}^k], \tag{1.5}$$

$$P^{C_n}_{(k^n)}(x; t, 0) = \mathfrak{J}_{2k+2}[t^{(k+1)y^2-jy}, \Lambda; \mathbb{Z}^k], \tag{1.6}$$

$$P^{C_n}_{(k^n)}(x; t, t) = \mathfrak{J}_{2k}[t^{ky^2-jy}, \Lambda; \mathbb{Z}^k_{\mathrm{even}}], \tag{1.7}$$

$$P^{BC_n}_{(k^n)}(x; \tau^2, -\tau, 0) = \mathfrak{J}_{2k+1}[(-1)^y \tau^{(2k+1)y^2-2jy}, \Lambda; \mathbb{Z}^k]. \tag{1.8}$$

At width zero the normalized value in (1.5) is $P^{B_n}_{\varnothing}(x; t, 1) = 1$.

These are, in order, Conjectures 1.2(a)–(c), 1.3(a)–(b), and 1.4 of [7]. The literal $k = 0$ assertion of Conjecture 1.2(c) would give $1/2$ on the right of (1.5); its correct normalized endpoint is therefore separate from the positive-width identity.

Theorem B. *For $n, k \ge 1$,*

$$P^{B_n}_{(k^n)}(x; t, t) = \tfrac12\mathfrak{J}_{2k}[(-1)^y t^{ky^2-(j-1)y}, \widehat{h}; \mathbb{Z}^k]. \tag{1.9}$$

The normalized width-zero value is $P^{B_n}_{\varnothing}(x; t, t) = 1$.

Theorem B is the additional, unnumbered identity in [7, Section 6]. Our Hall–Littlewood conventions are those of [7, (2.16)–(2.17)]; for the type-BC Weyl sum and polynomiality, see also [6, Definition 2.4 and Theorems 2.5–2.6].

The proof uses complementary minors, a familiar source of dual symmetric-function determinants [1, 4]. The corresponding proportionality of complementary maximal minors is also the basic Gale-dual relation; see [5, Lemma 2.10]. We apply it to the kernel pairs $\Phi D = 0$ and $V^{\mathsf{T}}C = 0$. Here $\det(\Phi V)/\Delta$ is a finite Weyl representation, while $D^{\mathsf{T}}C$ is the affine matrix. The coefficient kernels D come from (1.1); the affine kernels C come from reflections of weighted residue classes. Explicit complementary blocks fix all signs and constants.

Section 2 derives the finite Weyl determinants. Section 3 gives the normalized transfer and the three coefficient kernels. Section 4 treats four ordinary cases in two pairs. Section 5 treats the fixed residue, fixed center, and adjacent residues responsible for the three half-determinant formulas. Width zero and the classical specializations are collected in Section 6.

2 Finite Weyl determinants

We abbreviate the seven polynomials by $P^{\star}_{n,k}$, where $\star = C_0, C_t, BC, B_0, B_-, B_+, B_t$ denotes the type and boundary parameter in Theorems A and B. For $(a;t)_m = \prod_{r=0}^{m-1}(1-at^r)$, set

$$V_n(t) = \prod_{r=1}^{n}(1+t+\cdots+t^{r-1}).$$

The rectangular Weyl-sum definition, including width zero, is

$$P^{\star}_{n,k} = \frac{1}{V_n(t)\nu_{\star}(k)} \sum_{w\in W_n} w\left[\prod_{a=1}^{n} x_a^{-k}\rho_{\star}(x_a)\prod_{a<b}\frac{(tx_a-x_b)(1-tx_ax_b)}{(x_a-x_b)(1-x_ax_b)}\right], \tag{2.1}$$

where $W_n = S_n \ltimes \{\pm 1\}^n$ acts by permutations and inversions, $\nu_{\star}(k) = 1$ for $k \geq 1$, and the remaining data are as follows. In the BC and B_- rows, $t = \tau^2$.

Case $\star$	$\rho_{\star}(z)$	$\nu_{\star}(0)$
C_0	$(1-z^2)^{-1}$	1
C_t	$(1-tz^2)/(1-z^2)$	$(-t;t)_n$
BC	$(1+\tau z)/(1-z^2)$	1
B_0	$(1-z)^{-1}$	1
B_-	$(1+\tau z)/(1-z)$	$(\tau;\tau^2)_n$
B_+	1	$(-1;t)_n$
B_t	$(1-tz)/(1-z)$	$(-t;t)_n$

Indeed, the symmetric-group normalization of either (k^n), $k>0$, or $\varnothing$ is $V_n(t)$; only the boundary Pochhammer factor changes at width zero.

The standard symmetrizer identity

$$\sum_{\sigma\in S_n}\prod_{a<b}\frac{tz_{\sigma(a)}-z_{\sigma(b)}}{z_{\sigma(a)}-z_{\sigma(b)}} = V_n(t)$$

follows by alternation and comparison of degrees; setting $z_n = 0$ gives the recurrence $V_n = (1+t+\cdots+t^{n-1})V_{n-1}$. Since the other factors in (2.1) are symmetric, it reduces that expression to

$$P^{\star}_{n,k} = \frac{1}{\nu_{\star}(k)} \sum_{\eta\in\{\pm1\}^n}\prod_{a} u_a^{-k}\rho_{\star}(u_a)\prod_{a<b}\frac{1-tu_au_b}{1-u_au_b}, \qquad u_a = x_a^{\eta_a}. \tag{2.2}$$

2.1 Alternants and finite foldings

Write $N_t(z) = \prod_{a<b}(z_a-z_b)(1-tz_az_b)$ and put

$$f_1(z) = z^{n-1}, \qquad f_r(z) = z^{n-r} + t^{r-1}z^{n+r-2} \quad (2\leq r\leq n).$$

The finite numerator calculations reduce to three alternants.

Lemma 2.1. *With rows indexed by $1 \le a \le n$ and columns by $1 \le r \le n$,*

$$\det[f_r(z_a)] = N_t(z), \tag{2.3}$$

$$\det[z_a^{n-r} - t^r z_a^{n+r}] = \prod_a (1 - tz_a^2) N_t(z), \tag{2.4}$$

$$\det[z_a^{n-r} + \tau^{2r-1} z_a^{n+r-1}] = \prod_a (1 + \tau z_a) N_{\tau^2}(z). \tag{2.5}$$

Proof. Each determinant is alternating. On setting $z_b = (tz_a)^{-1}$, rows a, b are proportional, with multipliers $t^{1-n} z_a^{2-2n}$, $-t^{-n} z_a^{-2n}$, and $\tau^{1-2n} z_a^{1-2n}$, respectively; in the last expression $t = \tau^2$. This gives the factors $1 - tz_a z_b$. The second determinant has row factor $1 - tz_a^2$, and the third vanishes at $z_a = -\tau^{-1}$. Divisibility can be checked over the fraction field in the remaining variables and returned to the polynomial ring by Gauss's lemma. The factors are pairwise coprime and, together with $z_a - z_b$, have total degrees

$$3\binom{n}{2}, \qquad 2n + 3\binom{n}{2}, \qquad n + 3\binom{n}{2},$$

which are the respective upper bounds on the determinant degrees. In each case the lowest-degree homogeneous part is $\det[z_a^{n-r}] = \prod_{a<b}(z_a - z_b)$, fixing the constant at 1. □

Set $\phi_m(z) = z^m - z^{-m}$, $\psi_m(z) = z^m - z^{1-m}$, and $C_m(z) = z^m + z^{-m}$. For $L = n + k$ and standard basis $\mathbf{e}_1, \dots, \mathbf{e}_L$, use the foldings

$$[m]_C = \begin{cases} \mathbf{e}_m & m > 0, \\ 0 & m = 0, \\ -\mathbf{e}_{-m} & m < 0, \end{cases} \quad [m]_B = \begin{cases} \mathbf{e}_m & m \ge 1, \\ -\mathbf{e}_{1-m} & m \le 0, \end{cases}$$

$$[m]_+ = \begin{cases} \mathbf{e}_m & m \ge 2, \\ 2\mathbf{e}_1 & m = 1, \\ \mathbf{e}_{2-m} & m \le 0. \end{cases}$$

Only indices whose folded rows lie in $\{1, \dots, L\}$ will occur. The evaluation matrices are

$$\Phi^C_{a,p} = \phi_p(x_a), \qquad \Phi^B_{a,p} = \psi_p(x_a),$$

$$\Phi^+_{a,p} = \begin{cases} 1, & p = 1, \\ C_{p-1}(x_a), & p \ge 2. \end{cases}$$

Thus $\Phi^C[m]_C = \phi_m(x)$, $\Phi^B[m]_B = \psi_m(x)$, and $\Phi^+[m]_+ = C_{m-1}(x)$, with vector evaluation understood. In particular, $[1]_+ = 2\mathbf{e}_1$ is forced by $C_0 = 2$.

The three initial minors are the Weyl denominators

$$\begin{aligned} \Delta_C(z) &= \det[\phi_r(z_a)] = (-1)^n \prod_a z_a^{-n}(1 - z_a^2) N_1(z), \\ \Delta_B(z) &= \det[\psi_r(z_a)] = (-1)^n \prod_a z_a^{-(n-1)}(1 - z_a) N_1(z), \\ \Delta_+(z) &= \det[\chi_r(z_a)] = \prod_a z_a^{-(n-1)} N_1(z), \end{aligned} \tag{2.6}$$

where $\chi_1 = 1$ and $\chi_r = C_{r-1}$ for $r \geq 2$. These evaluations are immediate Vandermonde determinants in $z + z^{-1}$: $\phi_r/(z - z^{-1})$, $\psi_r/(z-1)$, and χ_r are monic polynomials of degree $r-1$. Inversion of a row multiplies the three denominators by -1, $-z^{-1}$, and 1, respectively.

2.2 The finite column matrices

For $\diamond \in \{C, B, +\}$, define

$$\begin{aligned} u_1^\diamond &= [k+1]_\diamond, \qquad \widetilde{u}_1^\diamond = [k]_\diamond, \\ u_r^\diamond &= [k+r]_\diamond + t^{r-1}[k+2-r]_\diamond \quad (r \geq 2), \\ \widetilde{u}_r^\diamond &= [k+r-1]_\diamond + t^{r-1}[k+1-r]_\diamond \quad (r \geq 2). \end{aligned}$$

The notation $\widetilde{u}_r^\diamond$ is used only for $\diamond = C, B$. Let $V^\star = (v_1^\star, \ldots, v_n^\star)$ be given by Table 1; every folded row there is at most $k+r$ in column r.

$\star$	Φ	$v_r^\star$	Δ
C_0	Φ^C	u_r^C	Δ_C
C_t	Φ^C	$[k+r]_C - t^r[k-r]_C$	Δ_C
BC	Φ^C	$u_r^C + \tau\widetilde{u}_r^C$	Δ_C
B_0	Φ^B	u_r^B	Δ_B
B_-	Φ^B	$[k+r]_B + \tau^{2r-1}[k+1-r]_B$	Δ_B
B_+	Φ^+	u_r^+	Δ_+
B_t	Φ^B	$u_r^B - t\widetilde{u}_r^B$	Δ_B

Table 1: Finite column data. In the BC and B_- rows, $t = \tau^2$.

Proposition 2.2. *For $n, k \geq 1$, the data of Table 1 satisfy*

$$P_{n,k}^\star = \frac{\det(\Phi V^\star)}{\Delta}.$$

Moreover, $\det V_H^\star = 1$ *for* $H = \{k+1, \ldots, k+n\}$.

Proof. Multiply (2.2) by the relevant denominator in (2.6). In the C_0 case, (2.3) gives row functions $z^{-n-k} f_r(z)$. Summing over inversions folds each row by $g(z) \mapsto g(z) - g(z^{-1})$, and these functions give $-\Phi^C u_r^C$. The n column signs cancel the factor $(-1)^n$ in Δ_C. The BC calculation multiplies the same row functions by $1 + \tau z$, giving $-\Phi^C(u_r^C + \tau\widetilde{u}_r^C)$. For C_t, replace f_r by $z^{n-r} - t^r z^{n+r}$ using (2.4); the folded column is $-\Phi^C([k+r]_C - t^r[k-r]_C)$.

For the shifted folding use $z^{-(n+k-1)} f_r(z)$ and $g(z) \mapsto g(z) - zg(z^{-1})$. This gives $-\Phi^B u_r^B$ in case B_0; multiplying by $1 - tz$ gives $-\Phi^B(u_r^B - t\widetilde{u}_r^B)$ in case B_t.

Equation (2.5) gives the B_- column in exactly the same way. Again the column signs cancel $(-1)^n$.

In case B_+, Δ_+ is invariant under each inversion. Use (2.3) and the row functions $z^{-(n+k-1)}f_r(z)$, now with $g(z) \mapsto g(z) + g(z^{-1})$. The first folded column is C_k, and the others are $C_{k+r-1} + t^{r-1}C_{k+1-r}$, exactly $\Phi^+ u_r^+$. This also checks the normalization of the first column without introducing an additional even-orthogonal alternant.

Finally, in every case the leading bracket places a 1 in row $k+r$ of column r. For $k \geq 1$ all other folded rows are strictly smaller, including those arising from nonpositive indices. Hence $V_H^\star$ is unit upper triangular. □

3 Complementary kernels and transfer

Throughout Sections 3–5, assume $n, k \geq 1$ and write

$$\begin{gathered} L = n + k, \qquad Q_0 = \{1, \dots, n\}, \qquad J = \{n+1, \dots, n+k\}, \\ H = \{k+1, \dots, k+n\}, \qquad I = \{1, \dots, k\}, \\ \omega = (-1)^{k(k+1)/2}, \qquad \zeta = (-1)^{nk}\omega. \end{gathered}$$

Subsets are always ordered increasingly. A subscript on an $n \times L$ matrix selects columns; on a matrix with L rows it selects rows.

3.1 The normalized transfer

Theorem 3.1. *Let F be a field, and let $X \in F^{n\times L}$, $Y \in F^{L\times n}$, and $U, Z \in F^{L\times k}$ satisfy $XU = 0$ and $Y^{\mathsf{T}}Z = 0$. If $\det X_{Q_0}$, $\det U_J$, $\det Y_H$, and $\det Z_I$ are nonzero, then*

$$\det(XY) = \frac{(-1)^{nk} \det X_{Q_0} \det Y_H}{\det U_J \det Z_I} \det(U^{\mathsf{T}} Z).$$

Proof. The nonzero minors give maximal rank, hence $\operatorname{col}(U) = \ker X$. The forms

$$\begin{aligned} (v_1, \dots, v_n) &\longmapsto \det[Xv_1 \ \cdots \ Xv_n], \\ (v_1, \dots, v_n) &\longmapsto \det[U \ v_1 \ \cdots \ v_n] \end{aligned}$$

induce nonzero top forms on $F^L/\operatorname{col}(U)$ and are therefore proportional. For $Q = \{q_1 < \cdots < q_n\}$ set $\epsilon(Q) = (-1)^{\sum_r (q_r - r)}$. Expansion along the last n columns gives

$$\det[U \ \mathbf{e}_{q_1} \ \cdots \ \mathbf{e}_{q_n}] = (-1)^{nk} \epsilon(Q) \det U_{Q^c}.$$

Consequently, for constants α, β independent of Q,

$$\det X_Q = \alpha \epsilon(Q) \det U_{Q^c}, \qquad \det Y_Q = \beta \epsilon(Q) \det Z_{Q^c},$$

where the second relation follows from $Y^{\mathsf{T}}Z = 0$. Cauchy–Binet gives

$$\det(XY) = \alpha\beta \sum_{|Q|=n} \det U_{Q^c} \det Z_{Q^c} = \alpha\beta \det(U^{\mathsf{T}} Z).$$

Evaluating the first relation at Q_0 and the second at H, for which $\epsilon(Q_0) = 1$ and $\epsilon(H) = (-1)^{nk}$, gives the result. □

In the finite representations, $\det \Phi_{Q_0} = \Delta$ and $\det V_H = 1$. Hence matrices D, C satisfying $\Phi D = 0$, $V^{\mathsf{T}} C = 0$, and having nonzero minors D_J, C_I give the single normalization formula

$$P^{\star}_{n,k} = \frac{(-1)^{nk}}{\det D_J \det C_I} \det(D^{\mathsf{T}} C). \tag{3.1}$$

Only the two indicated minors are needed in each application.

3.2 Three coefficient kernels

Palindromicity gives

$$\Lambda_i(-m) = -\Lambda_i(m), \qquad h_i(1-m) = h_i(m), \qquad h_i^+(2-m) = h_i^+(m). \tag{3.2}$$

In particular,

$$h_i(m) = \dot{e}_{n-i+m} + \dot{e}_{n-i-m+1}, \qquad h_i^+(m) = \dot{e}_{n-i+m} + \dot{e}_{n-i-m+2}.$$

Define $L \times k$ matrices by

$$D^C_{p,i} = (-1)^p \Lambda_i(p), \qquad D^B_{p,i} = (-1)^p h_i(p), \qquad D^+_{p,i} = (-1)^p h_i^+(p). \tag{3.3}$$

Lemma 3.2. *The three pairs satisfy* $\Phi^C D^C = 0$, $\Phi^B D^B = 0$, *and* $\Phi^+ D^+ = 0$. *Their complementary minors are* $\det D^C_J = \det D^B_J = \zeta$ *and* $\det D^+_J = 2\zeta$.

Proof. The generating identities are

$$\sum_m \Lambda_i(m) z^m = z^{-n} \dot{E}(z)(z^i - z^{-i}),$$
$$\sum_m h_i(m) z^m = z^{-n} \dot{E}(z)(z^i + z^{1-i}),$$
$$\sum_m h_i^+(m) z^m = z^{-n} \dot{E}(z)(z^i + z^{2-i}).$$

They vanish at $z = -x_a$. Pair the integer indices by $m \leftrightarrow -m$, $m \leftrightarrow 1-m$, and $m \leftrightarrow 2-m$, respectively. The first two paired sums are $\sum_{p\ge1}(-1)^p \Lambda_i(p)\phi_p(x_a)$ and $\sum_{p\ge1}(-1)^p h_i(p)\psi_p(x_a)$. In the third, the fixed index $m = 1$ gives, after division by x_a,

$$-h_i^+(1) + \sum_{p\ge2} (-1)^p h_i^+(p) C_{p-1}(x_a) = 0.$$

All three kernels vanish for $p > L$, giving the asserted kernel relations. In particular, the first row of D^+ has no extra factor 2.

For $1 \le r, i \le k$, the last k rows satisfy

$$D^C_{n+r,i} = D^B_{n+r,i} = (-1)^{n+r} \dot{e}_{i-r},$$
$$D^+_{n+r,i} = (-1)^{n+r} (\dot{e}_{i-r} + \dot{e}_{2-i-r}).$$

These blocks are upper triangular. The first two have diagonal $(-1)^{n+r}$; the third has the same diagonal except that its first entry is $2(-1)^{n+1}$, giving the stated minors. □

3.3 Folding and finite unfolding

We will fold scalar sequences $b_j(m)$, $m \in \mathbb{Z}$, into affine matrices. The three conventions are

$$\begin{aligned}
\mathcal{C}^C(b)_{p,j} &= b_j(p) - b_j(-p),\\
\mathcal{C}^B(b)_{p,j} &= (-1)^p\Big(b_j(p) + b_j(1-p)\Big),\\
\mathcal{C}^+(b)_{p,j} &= \begin{cases} -b_j(1), & p = 1,\\ (-1)^p\Big(b_j(p) + b_j(2-p)\Big), & 2 \le p \le L. \end{cases}
\end{aligned} \tag{3.4}$$

The sequences need not have finite support; the products below do.

Lemma 3.3. *For the matrices in* (3.3) *and* (3.4)*,*

$$((D^C)^{\mathsf{T}}\mathcal{C}^C(b))_{ij} = \sum_{m\in\mathbb{Z}} (-1)^m \Lambda_i(m) b_j(m), \tag{3.5}$$

$$((D^B)^{\mathsf{T}}\mathcal{C}^B(b))_{ij} = \sum_{m\in\mathbb{Z}} h_i(m) b_j(m), \tag{3.6}$$

$$((D^+)^{\mathsf{T}}\mathcal{C}^+(b))_{ij} = \sum_{m\in\mathbb{Z}} h_i^+(m) b_j(m). \tag{3.7}$$

Proof. For the first identity, pair $m = p$ with $m = -p$, using (3.2) and $\Lambda_i(0) = 0$. For the second, pair $p \ge 1$ with $1 - p \le 0$. For the third, the row $p = 1$ contributes $h_i^+(1)b_j(1)$, while $p \ge 2$ pairs with $2 - p \le 0$. Thus the fixed center is counted once, not twice. The vanishing for $p > L$ established above justifies each unfolding. □

A common bound controls all subsequent row sums. If $K > 0$ and $\mathcal{E}_r = 0$ outside $[0, N]$, a row with entries

$$\mathcal{E}_{n-i+j-Ky} + \sigma\mathcal{E}_{n+i+j-Ky-\delta}, \qquad 1 \le j \le k, \quad \delta \in \{0, 1, 2\},$$

vanishes unless

$$\left\lceil \frac{n - i + 1 - N}{K} \right\rceil \le y \le \left\lfloor \frac{n + i + k - \delta}{K} \right\rfloor.$$

Indeed, outside this interval every index is above N or below 0; here $2i - \delta \ge 0$. Take $N = 2n$, or $2n + 1$ for $\widehat{h}$. Thus, if $A_{ij} = \sum_y w_j(y) F_i(j - Ky)$, finite row multilinearity gives $\det A = \mathfrak{J}_K[w, F; \mathbb{Z}^k]$. The even-sublattice case will use a parity projection instead.

Every weight has exponent $\frac{K}{2}y^2 - (j - \eta)y$, with $\eta \in \{0, \frac{1}{2}, 1\}$, $K/2 \ge k - \eta$, and $j - \eta \ge 0$. It is nonnegative for every integer y. In case B_0 it is an integer because $y^2 - y$ is even; in the square-root cases twice this exponent is the exponent of τ. The resulting affine expressions are therefore polynomial in t or τ, even though reflection relations will be proved in the corresponding rational function fields.

4 The four ordinary affine kernels

Four specializations have no doubled complementary block. The two unshifted foldings share one reflection, and the two shifted foldings share another.

4.1 The pair C_0 and BC

Proposition 4.1. *Equations* (1.6) *and* (1.8) *hold for* $n, k \geq 1$.

Proof. Take $K = 2k + 2$ for C_0 and $K = 2k + 1$ for BC, with $t = \tau^2$ in the latter case. For $1 \leq j \leq k$ define

$$b_j(m) = \begin{cases} t^{Kq^2/2+jq}, & m = j + Kq, \quad q \in \mathbb{Z}, \\ 0, & m \not\equiv j \pmod{K}, \end{cases}$$
$$c_j(m) = b_j(m) - b_j(-m),$$

and let $C = \mathcal{C}^C(b)$. Shifting q gives $b_j(m + K) = t^{m+K/2} b_j(m)$, hence

$$c_j(-m) = -c_j(m), \qquad c_j(m) + t^{m-K/2} c_j(K - m) = 0. \tag{4.1}$$

The second relation follows by applying the quasiperiodicity to $b_j(K - m)$ and $b_j(m - K)$; it holds also outside the residue classes.

For C_0, the relation at $m = k + 1$ gives $c_j(k + 1) = 0$, and at $m = k + r$ it gives $c_j(k + r) + t^{r-1} c_j(k + 2 - r) = 0$. Since the odd extension agrees with the bracket $[m]_C$, these are precisely $(V^{C_0})^{\mathsf{T}} C = 0$. For BC, the first column gives $c_j(k + 1) + \tau c_j(k) = 0$. For $r \geq 2$, put $h = k + r$. The four terms in its finite column split as

$$(v_r^{BC})^{\mathsf{T}} C_{\bullet,j} = \Big(c_j(h) + \tau^{2h-K} c_j(K - h)\Big) + \tau\Big(c_j(h - 1) + \tau^{2(h-1)-K} c_j(K - h + 1)\Big) = 0.$$

Thus the second kernel relation holds in both cases.

Because $K > 2k$, the residues j and $-j$ are disjoint and the first k rows give $C_I = I_k$. By (3.5) and $y = -q$,

$$(D^C)^{\mathsf{T}} C = A \operatorname{diag}((-1)^1, \ldots, (-1)^k),$$
$$A_{ij} = \sum_{y \in \mathbb{Z}} (-1)^{Ky} t^{Ky^2/2 - jy} \Lambda_i(j - Ky).$$

The sign $(-1)^{Ky}$ is 1 for C_0 and $(-1)^y$ for BC. Since $\det D_J^C = \zeta = (-1)^{nk}\omega$, (3.1) gives $P_{n,k}^{\star} = \det A$. Finite row multilinearity proves the two identities. □

4.2 The pair B_0 and B_-

Proposition 4.2. *Equations* (1.3) *and* (1.4) *hold for* $n, k \geq 1$.

Proof. Take $K = 2k + 1$ for B_0 and $K = 2k$ for B_-, with $t = \tau^2$ in the latter case. Define

$$b_j(m) = \begin{cases} (-1)^{Ky} t^{Ky^2/2 - (j-1/2)y}, & m = j - Ky, \quad y \in \mathbb{Z}, \\ 0, & m \not\equiv j \pmod{K}, \end{cases}$$

and put

$$\gamma_j(m) = (-1)^m\Big(b_j(m) + b_j(1 - m)\Big), \qquad C = \mathcal{C}^B(b).$$

Changing y to $y-1$ gives $b_j(m+K) = (-1)^K t^{m+(K-1)/2} b_j(m)$. Applying this relation to the two reflected terms yields

$$\gamma_j(1-m) = -\gamma_j(m), \qquad \gamma_j(m) = -t^{m-(K+1)/2}\gamma_j(K+1-m). \tag{4.2}$$

For example, both $b_j(K+1-m)$ and $b_j(m-K)$ acquire the factor $(-1)^K t^{(K+1)/2-m}$ from $b_j(1-m)$ and $b_j(m)$. The parity in the definition of γ_j leaves the minus sign in (4.2) for either parity of K.

For B_0, the fixed index $m = k+1$ in the second reflection has $\gamma_j(k+1) = 0$; the remaining columns are killed by $\gamma_j(k+r) + t^{r-1}\gamma_j(k+2-r) = 0$. For B_-, it gives instead $\gamma_j(k+r) + \tau^{2r-1}\gamma_j(k+1-r) = 0$. The first reflection makes evaluation against $[m]_B$ agree with $\gamma_j(m)$, so $V^{\mathsf{T}}C = 0$ in both cases.

For $1 \le p, j \le k$, $b_j(p) = \delta_{p,j}$, whereas $b_j(1-p) = 0$ because $1 \le j+p-1 \le 2k-1 < K$. Thus $C_I = \operatorname{diag}((-1)^1, \dots, (-1)^k)$ and $\det C_I = \omega$. The common pairing (3.6) gives

$$((D^B)^{\mathsf{T}}C)_{ij} = A_{ij} := \sum_{y\in\mathbb{Z}} (-1)^{Ky} t^{Ky^2/2-(j-1/2)y} h_i(j-Ky).$$

Now (3.1) and $\det D_J^B = \zeta$ give $P^\star_{n,k} = \det A$. For odd K the affine sign is $(-1)^y$; for even K it is absent. Finite multilinearity proves the stated identities. □

5 Fixed classes and adjacent residues

The remaining cases have one doubled complementary minor. We treat separately its three sources: a fixed residue, a fixed center, and the overlap of adjacent residues.

5.1 The fixed residue in C_t

Proposition 5.1. *Equation* (1.7) *holds for* $n, k \ge 1$.

Proof. Set $K = 2k$ and define

$$b_j(m) = \begin{cases} (-1)^q t^{kq^2+jq}, & m = j+2kq, \quad q \in \mathbb{Z}, \\ 0, & m \not\equiv j \pmod{2k}. \end{cases}$$

Let $c_j(m) = b_j(m) - b_j(-m)$ and $C = \mathcal{C}^C(b)$. The quasiperiodicity is now $b_j(m+2k) = -t^{m+k} b_j(m)$, so the calculation leading to (4.1) gives

$$c_j(-m) = -c_j(m), \qquad c_j(m) = t^{m-k} c_j(2k-m).$$

At $m = k+r$ this annihilates the column $[k+r]_C - t^r[k-r]_C$; hence $(V^{C_t})^{\mathsf{T}}C = 0$.

Unlike the ordinary cases, the residues k and $-k$ coincide. Their contributions add rather than cancel:

$$c_k(k+2kq) = 2(-1)^q t^{kq(q+1)} \quad (q \in \mathbb{Z}).$$

For $j < k$ the two residues remain disjoint. Consequently, $C_I = \mathrm{diag}(1, \dots, 1, 2)$ and $\det C_I = 2$. This includes $k = 1$, when the block consists of the single entry 2.

Write

$$a_{ij}(y) = t^{ky^2 - jy} \Lambda_i(j - 2ky),$$
$$A^+_{ij} = \sum_y a_{ij}(y), \qquad A^-_{ij} = \sum_y (-1)^y a_{ij}(y).$$

The pairing (3.5), with $y = -q$, gives $(D^C)^{\mathsf{T}} C = A^- \,\mathrm{diag}((-1)^1, \dots, (-1)^k)$. Thus the fixed-residue minor and (3.1) yield $P^{C_t}_{n,k} = \frac{1}{2} \det A^-$.

It remains to identify the even-sublattice expression, not the full lattice sum. The parity projector and finite multilinearity give

$$\sum_{y \in \mathbb{Z}^k_{\mathrm{even}}} \det[a_{ij}(y_i)] = \tfrac{1}{2}\Big(\det A^+ + \det A^-\Big).$$

In the last column, $y \mapsto 1 - y$ preserves $ky^2 - ky$ and sends $k - 2ky$ to its negative. Since Λ_i is odd, $a_{i,k}(1 - y) = -a_{i,k}(y)$. This fixed-point-free involution makes the last column of A^+ zero. The parity projection therefore equals $\frac{1}{2} \det A^-$, proving the identity. □

5.2 The fixed center in B_+

The sequence used here will also supply the adjacent-residue reflection in the next subsection. For $1 \le j \le k$, set

$$b_j(m) = \begin{cases} (-1)^y t^{ky^2 - (j-1)y}, & m = j - 2ky, \quad y \in \mathbb{Z}, \\ 0, & m \not\equiv j \pmod{2k}, \end{cases} \tag{5.1}$$
$$\Gamma_j(m) = (-1)^m \Big(b_j(m) + b_j(2 - m)\Big).$$

Lemma 5.2. *For every integer m,*

$$b_j(m + 2k) = -t^{m+k-1} b_j(m),$$
$$\Gamma_j(2 - m) = \Gamma_j(m),$$
$$\Gamma_j(m) = -t^{m-k-1} \Gamma_j(2k + 2 - m). \tag{5.2}$$

In particular, $\Gamma_j(k + 1) = 0$.

Proof. The first relation follows from $y \mapsto y - 1$ in (5.1); the second is the defining symmetry. For the last, the first relation gives

$$b_j(2k + 2 - m) = -t^{k+1-m} b_j(2 - m), \qquad b_j(m - 2k) = -t^{k+1-m} b_j(m).$$

Since m and $2k + 2 - m$ have the same parity, adding these relations proves (5.2). Its fixed index $m = k + 1$ has eigenvalue -1 and hence vanishes. □

Proposition 5.3. *Equation* (1.5) *holds for* $n, k \geq 1$.

Proof. Take $C = \mathcal{C}^+(b)$ for (5.1). The normalized center row is $C_{1,j} = -b_j(1)$, so

$$[1]_+^{\mathsf{T}} C_{\bullet,j} = 2C_{1,j} = -2b_j(1) = \Gamma_j(1).$$

Together with $\Gamma_j(2-m) = \Gamma_j(m)$, this shows that every bracket $[m]_+$ evaluates to $\Gamma_j(m)$, including the fixed center. Lemma 5.2 therefore gives $(V^{B_+})^{\mathsf{T}} C = 0$: the first column evaluates to $\Gamma_j(k+1)$ and the others to $\Gamma_j(k+r)+t^{r-1}\Gamma_j(k+2-r) = 0$.

The first k rows are $C_I = \operatorname{diag}((-1)^1, \ldots, (-1)^k)$. Indeed, $b_j(1) = \delta_{j,1}$; for $2 \leq p \leq k$, $b_j(p) = \delta_{p,j}$ and the reflected congruence would require $2ky = j+p-2 \in \{1, \ldots, 2k-2\}$, which is impossible. Thus $\det C_I = \omega$. The factor 2 occurs instead in $\det D_J^+ = 2\zeta$, whose first diagonal coefficient is $\dot{e}_0 + \dot{e}_0$ by Lemma 3.2.

By (3.7),

$$((D^+)^{\mathsf{T}} C)_{ij} = A_{ij}^{B_+} := \sum_y (-1)^y t^{ky^2-(j-1)y} h_i^+(j-2ky).$$

In this pairing the center contributes exactly $h_i^+(1)b_j(1)$; its coefficient is not doubled. Formula (3.1) now gives $P_{n,k}^{B_+} = \frac{1}{2}\det A^{B_+}$, and finite row multilinearity proves the result. □

5.3 Adjacent residues in B_t

This case uses the same coefficient matrix D^B as B_0 and B_-. Its different row kernel satisfies $\widehat{h}_i(m) = h_i(m) + h_i(m-1)$, directly from $\ddot{e}_r = \dot{e}_r + \dot{e}_{r-1}$. Keep b_j and Γ_j from (5.1) and put

$$\begin{aligned} a_j(m) &= b_j(m) + b_j(m+1), \\ c_j(m) &= (-1)^m \Big(a_j(m) + a_j(1-m) \Big), \\ \gamma_j(m) &= c_j(m) - t c_j(m-1). \end{aligned}$$

Thus a_j is supported on the two adjacent residues j and $j-1$, and $c_j(1-m) = -c_j(m)$.

Lemma 5.4. *The adjacent-residue sequences satisfy*

$$c_j(m) = \Gamma_j(m) - \Gamma_j(m+1), \tag{5.3}$$

$$\gamma_j(m) = (1+t)\Gamma_j(m) - \Gamma_j(m+1) - t\Gamma_j(m-1), \tag{5.4}$$

$$\gamma_j(m) = -t^{m-k-1}\gamma_j(2k+2-m). \tag{5.5}$$

In particular, $\gamma_j(k+1) = 0$.

Proof. Expanding the definitions proves (5.3), and subtracting $tc_j(m-1)$ gives (5.4). Let $T = 2k+2$ and $d = m-k-1$. Apply (5.2) at $m, m+1, m-1$ to the latter identity:

$$\begin{aligned} \gamma_j(m) &= -t^d \Big((1+t)\Gamma_j(T-m) - t\Gamma_j(T-m-1) - \Gamma_j(T-m+1) \Big) \\ &= -t^d \gamma_j(T-m). \end{aligned}$$

The fixed-index assertion follows at $m = k+1$. □

Proposition 5.5. *Equation* (1.9) *holds for* $n, k \geq 1$.

Proof. Set $C = \mathcal{C}^B(a)$, so $C_{p,j} = c_j(p)$. Evaluation against the first finite column gives

$$([k+1]_B - t[k]_B)^{\mathsf{T}} C_{\bullet,j} = \gamma_j(k+1) = 0.$$

For $r \geq 2$, the four terms in $u_r^B - t\tilde{u}_r^B$ give $\gamma_j(k+r) + t^{r-1}\gamma_j(k+2-r) = 0$ by (5.5). Hence $(V^{B_t})^{\mathsf{T}} C = 0$.

The first complementary block contains the overlap that fixes the normalization. For $1 \leq p, j \leq k$, only $y = 0$ contributes, and

$$a_j(p) = \delta_{p,j} + \delta_{p,j-1}, \qquad a_j(1-p) = \delta_{p,1}\delta_{j,1}.$$

The second equality follows because the two congruences require $2ky = j + p - 1$ or $2ky = j + p - 2$; only the latter can occur, at $p = j = 1$. Therefore

$$C_{p,j} = (-1)^p\Big(\delta_{p,j} + \delta_{p,j-1} + \delta_{p,1}\delta_{j,1}\Big) \quad (1 \leq p, j \leq k).$$

This block is upper bidiagonal, with diagonal $-2, 1, -1, 1, \ldots$, and hence $\det C_I = 2\omega$. There is no doubling in $\det D_J^B = \zeta$.

Finally, the common pairing (3.6) and the definition of $\widehat{h}_i$ give

$$\begin{aligned}((D^B)^{\mathsf{T}} C)_{ij} &= \sum_m h_i(m)\Big(b_j(m) + b_j(m+1)\Big) \\ &= \sum_y (-1)^y t^{ky^2-(j-1)y}\Big(h_i(j-2ky) + h_i(j-1-2ky)\Big) \\ &= \sum_y (-1)^y t^{ky^2-(j-1)y}\widehat{h}_i(j-2ky).\end{aligned}$$

The normalization formula yields one half of this determinant, and finite multilinearity proves Theorem B. □

6 Width zero and classical specializations

6.1 The normalized empty rectangle

Proposition 6.1. *For every* $n \geq 1$, *all seven normalized width-zero values are* 1. *Consequently, the five non-prefactored identities in Theorem A hold also at* $k = 0$.

Proof. The symmetric-group reduction (2.2) is valid at $k = 0$. The alternant and row-folding calculation in Proposition 2.2 therefore gives

$$\nu_\star(0) P^\star_{n,0} = \frac{\det(\Phi V^\star|_{k=0})}{\Delta},$$

without using the complementary-kernel transfer. Relative to the initial basis defining Δ, these column transformations are upper triangular. Their diagonal coefficients in column r are

Case	Diagonal coefficient	Product
C_0, BC, B_0	1	1
C_t, B_t	$1+t^r$	$(-t;t)_n$
B_-	$1-\tau^{2r-1}$	$(\tau;\tau^2)_n$
B_+	$1+t^{r-1}$	$(-1;t)_n$

For example, the B_t columns are $(1+t)\psi_1$ and

$$(1+t^r)\psi_r - (t+t^{r-1})\psi_{r-1} \qquad (r \geq 2),$$

while the BC columns are ϕ_1 and

$$\phi_r + (\tau - \tau^{2r-1})\phi_{r-1} - \tau^{2r-2}\phi_{r-2} \qquad (r \geq 2).$$

For B_+ the first column is $C_0 = 2$ relative to $\chi_1 = 1$, so its diagonal is indeed $1 + t^0 = 2$. Each product in the table equals $\nu_\star(0)$, proving the claim. The five affine expressions without an external half-factor are 1 by the empty-determinant convention. □

These are rational-function identities before cancellation; any removable singularities of the displayed normalization factors are understood after cancellation. In particular, the conclusion does not require a nonvanishing numerical value of every Pochhammer factor.

6.2 The specialization $t = 0$

The classical rectangular dual Jacobi–Trudi formulas are recovered in three finite foldings. They are the symplectic and orthogonal determinants in [2, Theorem 2.1]; see also the lattice-path proofs in [3].

Corollary 6.2. *For $n, k \geq 1$, with $1 \leq a, r \leq n$ on the left and $1 \leq i, j \leq k$ on the right,*

$$\frac{\det[\phi_{k+r}(x_a)]}{\Delta_C(x)} = \det[\dot{e}_{n-i+j} - \dot{e}_{n+i+j}], \tag{6.1}$$

$$\begin{aligned}\frac{\det[\psi_{k+r}(x_a)]}{\Delta_B(x)} &= \det[\dot{e}_{n-i+j} + \dot{e}_{n+i+j-1}] \\ &= \tfrac{1}{2}\det[\ddot{e}_{n-i+j} + \ddot{e}_{n+i+j-1}],\end{aligned} \tag{6.2}$$

$$\frac{\det[C_{k+r-1}(x_a)]}{\Delta_+(x)} = \tfrac{1}{2}\det[\dot{e}_{n-i+j} + \dot{e}_{n+i+j-2}]. \tag{6.3}$$

The seven finite representations specialize to the corresponding left sides: C_0, C_t, BC to (6.1); B_0, B_-, B_t to (6.2); and B_+ to (6.3).

Proof. At $t = 0$ (and $\tau = 0$ in the square-root cases), every finite column in Table 1 reduces to $[k + r]_\diamond$. This identifies the finite ratios. In the affine formulas for C_0, B_0, B_+, B_t, every nonzero shift has a strictly positive parameter exponent, leaving only $y = 0$ in each row and giving the displayed dual determinants. For C_t one should instead use its finite ratio: at $j = k, y = 1$ the weight exponent is zero, so nonzero shifts do not disappear entrywise. The already proved parity identity identifies its specialized even-sublattice sum with the same symplectic determinant. □

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